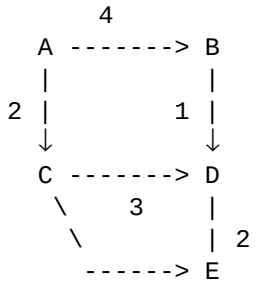
Bellman-Ford Algorithm — Step-by-Step Example

Find the shortest distance from source A to all other vertices.

Question

Consider the following directed weighted graph. Apply Bellman-Ford with source A.



Edges: A→B=4, A→C=2, B→D=1, C→D=3, C→E=3, D→E=2.

There are 5 vertices, so V−1 = 4 passes are required.

Step 1 — INITIALIZE-SINGLE-SOURCE

Vertex	d[v]	π[v]
A	0	—
B	∞	—
C	∞	—
D	∞	—
E	∞	—

PASS 1 — Relax every edge

```
1. A→B (4):  ∞ > 0+4  → d[B]=4
2. A→C (2):  ∞ > 0+2  → d[C]=2
3. B→D (1):  ∞ > 4+1  → d[D]=5
4. C→D (3):  5 > 2+3 → FALSE, no change
5. C→E (3):  ∞ > 2+3  → d[E]=5
6. D→E (2):  5 > 5+2 → FALSE, no change
```

After Pass 1: A=0, B=4, C=2, D=5, E=5

PASS 2 — Relax every edge again

```
A→B: 4 > 0+4 → no change
A→C: 2 > 0+2 → no change
B→D: 5 > 4+1 → no change
C→D: 5 > 2+3 → no change
C→E: 5 > 2+3 → no change
D→E: 5 > 5+2 → no change
```

After Pass 2: A=0, B=4, C=2, D=5, E=5

PASS 3 — Relax every edge again

```
No distance can be improved.
```

After Pass 3: A=0, B=4, C=2, D=5, E=5

PASS 4 — Relax every edge again

```
No distance can be improved.
```

After Pass 4: A=0, B=4, C=2, D=5, E=5

Final Negative-Cycle Check

Now check every edge one more time. If any edge still satisfies d[v] > d[u] + w(u,v), return FALSE; otherwise return TRUE.

```
A→B: 4 > 0+4  → FALSE
A→C: 2 > 0+2  → FALSE
B→D: 5 > 4+1  → FALSE
C→D: 5 > 2+3  → FALSE
C→E: 5 > 2+3  → FALSE
D→E: 5 > 5+2  → FALSE

No edge can be relaxed.
Therefore: return TRUE.
There is NO reachable negative-weight cycle.
```

Final Answer

Vertex	Shortest Distance	Shortest Path
A	0	A
B	4	A → B
C	2	A → C
D	5	A → B → D
E	5	A → C → E

Key Point

Bellman-Ford: Initialize → Relax ALL edges V−1 times → Check ALL edges one extra time for a possible negative-weight cycle.